MILITARY SPECIFICATION SHEET

FILTERS, BAND PASS, TELEMETRY, TYPE FR7RX22ZZ1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the filters described herein shall consist of this document and the latest issue of Specification MIL-F-18327.

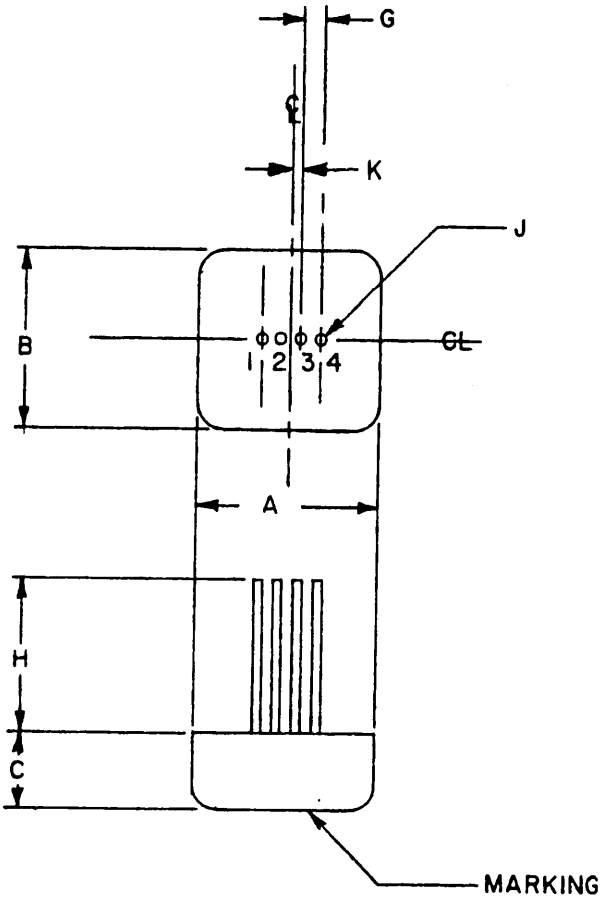


FIGURE 1. Case and mounting dimensions.

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Type designation	Part number M18327/046	Dimensions A	B	C	G (Typ)	H	J (Dia. 4 reqd)	Weight (oz)	K
FR7RX22ZZ1	001	1.219(30.96) 1.157(29.39)	1.219(30.96) 1.157(29.39)	.531(13.49) .406(10.31)	.210(5.33) .190(4.83)	1.125(28.58) .875(22.22)	.025(.64)	1	.105(2.67) .095(2.41)
	002	.734(18.64) .672(17.07)	.734(18.64) .672(17.07)	.531(13.49) .406(10.31)	.110(2.79) .090(2.29)			.25	.055(1.40) .045(1.14)
	003	.734(18.64) .672(17.07)	.734(18.64) .672(17.07)	.36 (9.1) .30 (7.6)	.110(2.79) .090(2.29)			.20	.055(1.40) .045(1.14)
	004	1.219(30.96) 1.157(29.39)	1.219(30.96) 1.157(29.39)	.531(13.49) .406(10.31)	.210(5.33) .190(4.83)				.105(2.67) .095(2.41)
	005								
	006								
	007								
	008								
	009								
	010								
	011								
	012								
	013								
	014	1.219(30.96) 1.157(29.39)	1.219(30.96) 1.157(29.39)		.210(5.33) .190(4.83)			.20	
	015	.734(18.64) .672(17.07)	.734(18.64) .672(17.07)		.110(2.79) .090(2.29)			.33	.055(1.40) .045(1.14)
	016								
	017								
	018								
	019								
	020								
	021								
	022								
	023								
	024			.531(13.49) .406(10.31)				.33	
	025			.36 (9.1) .30 (7.6)				.20	
	026								
	027								
	028			.36 (9.1) .30 (7.6)				.20	
	029			.531(13.49) .406(10.31)				.33	
	030								
	031								
	032								
	033								
	034			.36 (9.1) .30 (7.6)				.20	
	035								
	036								

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
4. Terminal identification: 1-2 input; 3-4 output, these filters are electrically symmetrical and may be installed in reverse position without changing the performance characteristics.
5. Where values are written one over the other, they represent maximum and minimum values respectively.

FIGURE 1. Case and mounting dimensions - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case: Metal (5 sides) with terminals side encapsulated.

Terminals: Solderable pin.

Weight: See figure 1.

Terminal strength: Method 211, MIL-STD-202. Test condition A (pull), 2 pounds.

Solderability: Method 208, MIL-STD-202. All terminations of each filter to be tested.
Solder dip dwell time: 5 seconds ±1/2 second.
Examination of terminations: At 10X magnification.

Dielectric withstanding voltage:
Terminals to case:
At atmospheric pressure: Method 301, MIL-STD-202. 500 Vrms between each terminal and case.
At reduced pressure: Method 301, MIL-STD-202, and Method 105, test condition B, MIL-STD-202. 300 Vrms between each terminal and case.

Input to output: Method 301, MIL-STD-202. The following details shall apply:
Magnitude and nature of test voltage: 50 Vdc.
Duration of application of test voltage: 5 ± 1 second. The test voltage shall be raised from 0 to the specified value within 1 second maximum.
Points of application of test voltage: Between terminals 1 and 2 tied together and 3 and 4 tied together. Test shall be repeated with reverse voltage polarity.
Limiting of surge current: 30 to 50 mA.
Examinations after test: Filters shall be examined for evidence of breakdown and damage.

Insulation resistance: Method 302, test condition B, MIL-STD-202. Not less than 10,000 megohms when measured between each terminal and case.

Stability at temperature extremes:
1 hour at -55° +0°, -3°C.
1 hour at +105° +3°, -0°C.

Measurements at 25°C after test:
Dielectric withstanding voltage - At atmospheric pressure.
Electrical characteristics - As specified in table I.

Electrical characteristics: See table I.

DC operating voltage: Zero volts.

Life: Method 108, test condition F, MIL-STD-202.
Test temperature 105° +10°, -5°C.
Signal input level - See table I.
Measurements at 25°C after test:
Dielectric withstanding voltage - At atmospheric pressure.
Insulation resistance - Not less than 1,000 megohms between each terminal and case.
Electrical characteristics - As specified in table I.

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Temperature rise: Not applicable.

Vibration, high frequency: Method 204, test condition B, MIL-STD-202.
Electrical load not applicable.
Measurements at 25°C after test:
Dielectric withstanding voltage - At atmospheric pressure.
Electrical characteristics - As specified in table I.

Shock (specified pulse): Method 213, test condition I, MIL-STD-202.
Measurements at 25°C after test:
Dielectric withstanding voltage - At atmospheric pressure.
Electrical characteristics - As specified in table I.

Thermal shock: Method 107, test condition A, MIL-STD-202, except that the temperature at step 3 shall be 105°C.
Exposure time - 30 minutes.
Measurements at 25°C after test:
Dielectric withstanding voltage - At atmospheric pressure.
Insulation resistance - Not less than 10,000 megohms between each terminal and case.

Immersion: Method 104, test condition A, MIL-STD-202. After final cycle, filters shall be washed under running tap water and dried. After the drying period, the filters shall be subjected to the following:
Measurements at 25°C.
Dielectric withstanding voltage - At atmospheric pressure.
Insulation resistance - Not less than 10,000 megohms between each terminal and case.

Moisture resistance: Method 106, MIL-STD-202.
Measurements at 25°C after test:
Dielectric withstanding voltage - At atmospheric pressure.
Insulation resistance - Not less than 10,000 megohms between each terminal and case.

Flammability: Method 111, MIL-STD-202. Duration of visible flame shall not exceed 3 minutes after removal of the applied flame. There shall be no explosion nor any violent burning which results in an explosive type flame. There shall be no dripping of flaming material from the filter under test.

Part number: M18327/046-(dash number from table I).

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TABLE I. Electrical characteristics.

Type designation	Dash number	Impedance (ohms) Source	Impedance (ohms) Load	Reference frequency (Hz)	Insertion loss (dB)	Frequency range (Hz)	Discrimination (dB) Min	Discrimination (dB) Max	Signal input level (volts to source resistor)	IRIG STD-106 Channel 1/
FR7RX22ZZ1	001	10K	10K	400	8 max	228	40		2.0	1
						300	15			
						370		4		
						430		4		
						500	15			
						700	40			
	002			7,350	8 max	4,190	40			11
						5,512	15			
						6,799		4		
						7,901		4		
						9,188	15			
						12,863	40			
	003			93,000	6 max	53,940	40			10
						69,750	15			
						80,025		4		
						99,975		4		
						115,250	15			
						162,750	40			
	004			560	8 max	325	40			2
						420	15			
						518		4		
						602		4		
						700	15			
						980	40			
	005			730	8 max	423	40			3
						548	15			
						675		4		
						785		4		
						913	15			
						1,278	40			
	006			960	6 max	557	40			4
						720	15			
						888		4		
						1,032		4		
						1,200	15			
						1,680	40			
	007			1,000	6 max	580	40			N/A
						750	15			
						925		4		
						1,075		4		
						1,250	15			
						1,750	40			
	008			1,300	6 max	754	40			5
						975	15			
						1,202		4		
						1,398		4		
						1,625	15			
						2,275	40			

See footnote at end of table.

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TABLE I. Electrical characteristics - Continued.

Type designation	Dash number	Impedance (ohms) Source	Impedance (ohms) Load	Reference frequency (Hz)	Insertion loss (dB)	Frequency range (Hz)	Discrimination (dB) Min	Discrimination (dB) Max	Signal input level (volts to source resistor)	IRIG STD-106 Channel 1/
FR7RX22ZZ1	009	10K	10K	1,700	6 max	986	40		2.0	6
						1,275	15			
						1,572		4		
						1,828		4		
						2,125	15			
						2,975	40			
	010			2,300		1,334	40			7
						1,725	15			
						2,127		4		
						2,473		4		
						2,875	15			
						4,025	40			
	011			3,000		1,740	40			8
						2,250	15			
						2,775		4		
						3,225		4		
						3,750	15			
						5,250	40			
	012			3,900		2,262	40			9
						2,925	15			
						3,607		4		
						4,193		4		
						4,875	15			
						6,825	40			
	013			4,000		2,320	40			N/A
						3,000	15			
						3,700		4		
						4,300		4		
						5,000	15			
						7,000	40			
	014			5,400		3,132	40			10
						4,050	15			
						4,995		4		
						5,805		4		
						6,750	15			
						9,450	40			
	015			8,000		4,640	40			N/A
						6,000	15			
						7,400		4		
						8,600		4		
						10,000	15			
						14,000	40			
	016			10,500		6,090	40			12
						7,875	15			
						9,712		4		
						11,288		4		
						13,125	15			
						18,375	40			

See footnote at end of table.

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TABLE I. Electrical characteristics - Continued.

Type designation	Dash number	Impedance (ohms) Source	Impedance (ohms) Load	Reference frequency (Hz)	Insertion loss (dB)	Frequency range (Hz)	Discrimination (dB) Min	Discrimination (dB) Max	Signal input level (volts to source resistor)	IRIG STD-106 Channel 1/
FR7RX22ZZ1	017	10K	10K	12,000	6 max	6,960	40		2.0	N/A
						9,000	15			
						11,100		4		
						12,900		4		
						15,000	15			
						21,000	40			
	018			14,500		8,410	40			13
						10,875	15			
						13,412		4		
						15,588		4		
						18,125	15			
						25,375	40			
	019			22,000		12,760	40			14
						16,500	15			
						20,350		4		
						23,650		4		
						27,500	15			
						38,500	40			
	020			30,000		17,400	40			15
						22,500	15			
						27,750		4		
						32,250		4		
						37,500	15			
						52,500	40			
	021			40,000		23,200	40			16
						30,000	15			
						37,000		4		
						43,000		4		
						50,000	15			
						70,000	40			
	022			52,500		30,450	40			17
						39,375	15			
						48,562		4		
						56,438		4		
						65,625	15			
						91,875	40			
	023			56,000		32,480	40			6A
						42,000	15			
						51,800		4		
						60,200		4		
						70,000	15			
						98,000	40			
	024			70,000		40,600	40			18
						52,500	15			
						64,750		4		
						75,250		4		
						87,500	15			
						122,500	40			

See footnote at end of table.

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TABLE I. Electrical characteristics - Continued.

Type designation	Dash number	Impedance (ohms)		Reference frequency (Hz)	Insertion loss (dB)	Frequency range (Hz)	Discrimination (dB)		Signal input level (volts to source resistor)	IRID STD-106 Channel 1/
		Source	Load				Min	Max		
FR7RX22ZZ1	025	10K	10K	96,000	6 max	55,680	40		2.0	11A
						72,000	15			
						88,800		4		
						103,200		4		
						120,000	15			
						168,000	40			
	026			124,000		71,920	40			20
						93,000	15			
						114,700		4		
						133,300		4		
						155,000	15			
						217,000	40			
	027			160,000		92,800	40			19A
						120,000	15			
						148,000		4		
						172,000		4		
						200,000	15			
						280,000	40			
	028			165,000		95,700	40			21
						123,750	15			
						152,624		4		
						177,375		4		
						206,250	15			
						288,750	40			
	029			22,000	3 max	55,000	40			A
						33,000	15			
						25,300		4		
						18,700		4		
						11,000	15			
						8,800	40			
	030			30,000		75,000	40			B
						45,000	15			
						34,500		4		
						25,500		4		
						15,000	15			
						12,000	40			
	031			40,000		100,000	40			C
						60,000	15			
						46,000		4		
						34,000		4		
						20,000	15			
						10,000	40			
	032			52,500		131,250	40			D
						78,750	15			
						60,375		4		
						44,625		4		
						26,250	15			
						21,000	40			

See footnote at end of table.

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TABLE I. Electrical characteristics - Continued.

Type designation	Dash number	Impedance (ohms)		Reference frequency (Hz)	Insertion loss (dB)	Frequency range (Hz)	Discrimination (dB)		Signal input level (volts to source resistor)	IRIG STD-106 Channel 1/
		Source	Load				Min	Max		
FR7RX22ZZ1	033	10K	10K	70,000	3 max	175,000	40		2.0	E
						105,000	15			
						80,500		4		
						59,500		4		
						35,000	15			
						28,000	40			
	034			93,000		205,500	40			F
						139,500	15			
						106,950		4		
						79,050		4		
						46,500	15			
						37,200	40			
	035			124,000		310,000	40			G
						186,000	15			
						142,600		4		
						105,400		4		
						62,000	15			
						49,600	40			
	036			165,000		412,500	40			H
						247,500	15			
						189,750		4		
						124,250		4		
						82,500	15			
						66,000	40			

1/ FM subcarrier channels in accordance with MIL-STD-1572, Telemetry Standards, for information only.

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QUALITY ASSURANCE PROVISIONS

Qualification of filters based on similarity to qualified filters. Qualification by similarity may be granted to all parts covered in this specification based on the qualification of one part number. Qualification by similarity shall be restricted to those parts which meet the criteria for a similar filter as specified in MIL-F-18327. The contractor shall provide certification that any parts qualified on the basis of similarity meet the full requirements of this specification and MIL-F-18327.

NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

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Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
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Air Force - 11, 17, 99
DLA - ES

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Army - ME
Navy - MC
Air Force - 19

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